

About (GitHub Sidebar)

HR Attrition & Workforce Intelligence Data Analytics Project. Analyzes 1,470 employee records using Excel, SQL, Python, and Power BI to identify turnover drivers and build a retention watchlist.

HR Attrition & Workforce Intelligence

Built an end-to-end data analytics project using Excel, SQL, Python, and Power BI to analyze HR data for 1,470 employees and find out why people leave the company.

The project uses a 35-column dataset to figure out what causes voluntary attrition and creates a watchlist to predict who might leave next.

How it was done:

- **Excel:** Checked data quality and used pivot tables to find that overtime workers have nearly 3x higher turnover.
- **SQL (MySQL):** Created a rule-based risk model that groups employees into High, Medium, and Low risk segments, showing a huge difference in turnover rates between segments.
- **Python:** Used Pandas, Seaborn, and Matplotlib to calculate correlations and build a 0-to-3 point risk score that accurately maps to actual attrition (from 5.7% up to 59.0%).
- **Power BI:** Built a 3-page interactive dashboard to summarize the data, drill down into specific job roles, and show a live watchlist of at-risk employees.

Key Findings

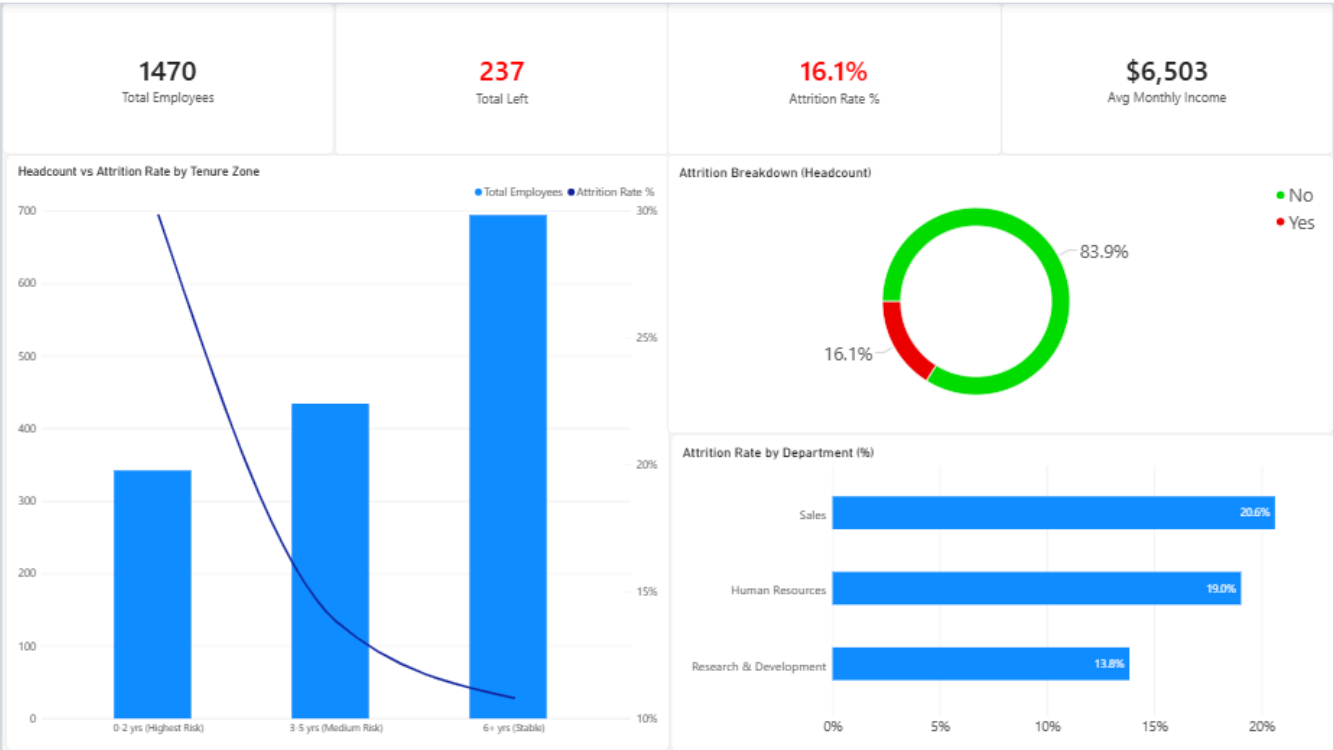
- **Overall Attrition:** 16.1% of employees (237 out of 1,470) left the company.
- **The Overtime Effect:** Employees working overtime have a 30.5% attrition rate, compared to 10.4% for those who don't.
- **Tenure Matters:** About 30% of employees leave within their first 2 years. This drops to ~17% for years 3-5 and ~11% after 6 years.

- **Pay Gap:** Employees who left made an average of \$4,787/month, while those who stayed made \$6,833/month (a \$2,046 difference).
- **High-Risk Roles:** Sales Representatives (39.8% attrition) and Laboratory Technicians (23.9% attrition) are leaving the most.
- **Watchlist:** Identified 153 current employees who are at high risk of leaving based on the data.

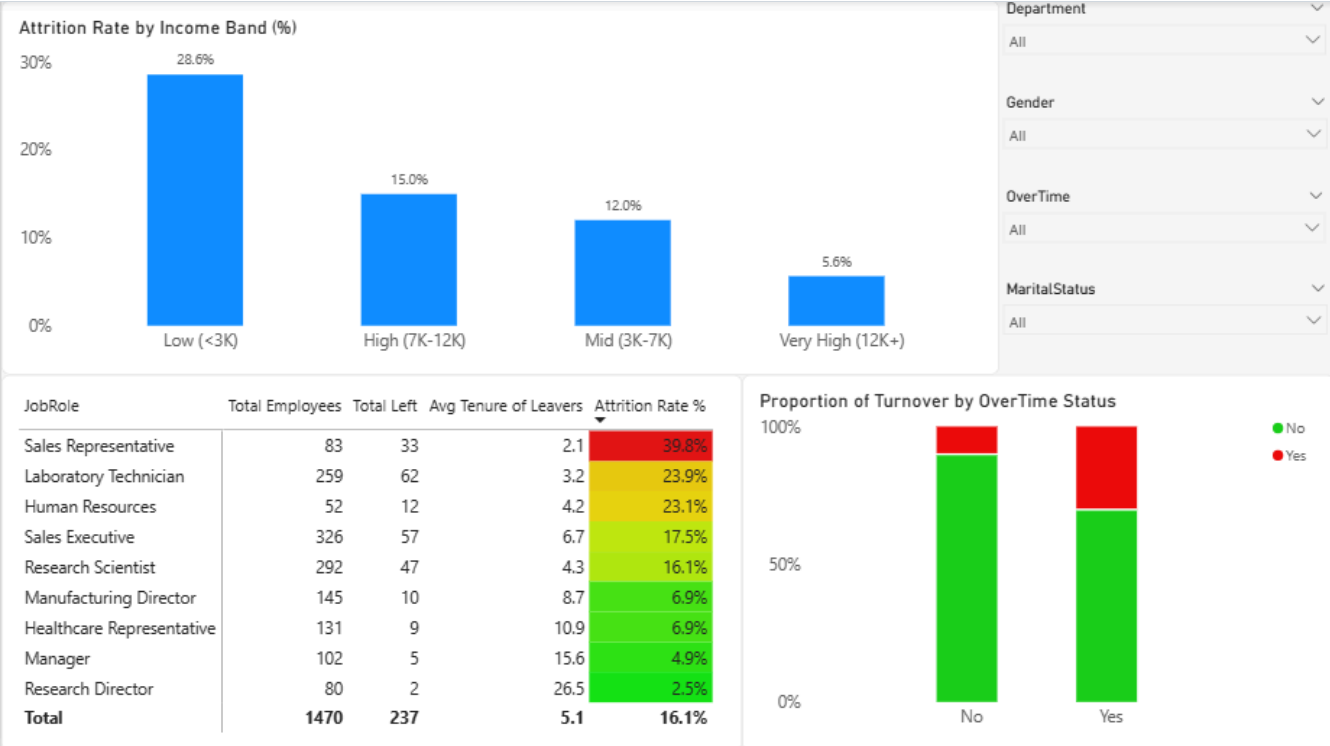
Power BI Dashboard

The project includes an interactive 3-page Power BI report.

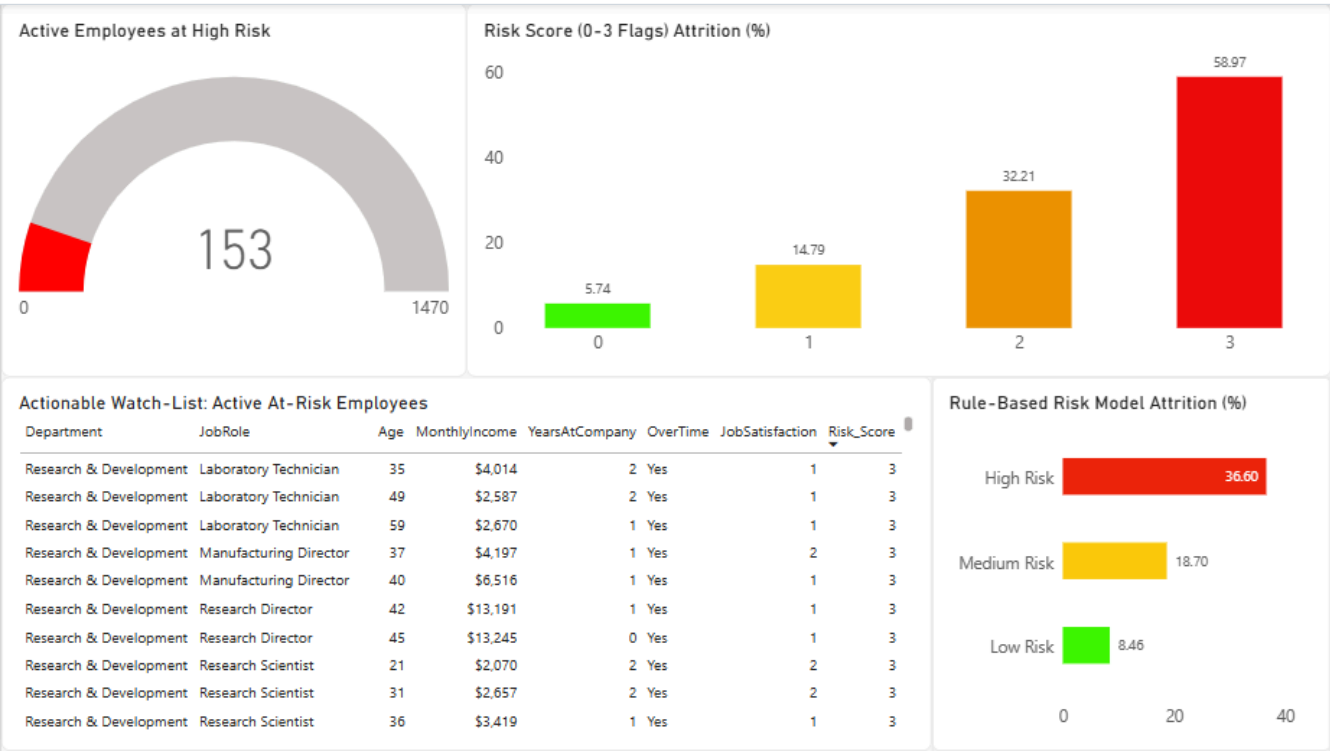
Page 1: Overview



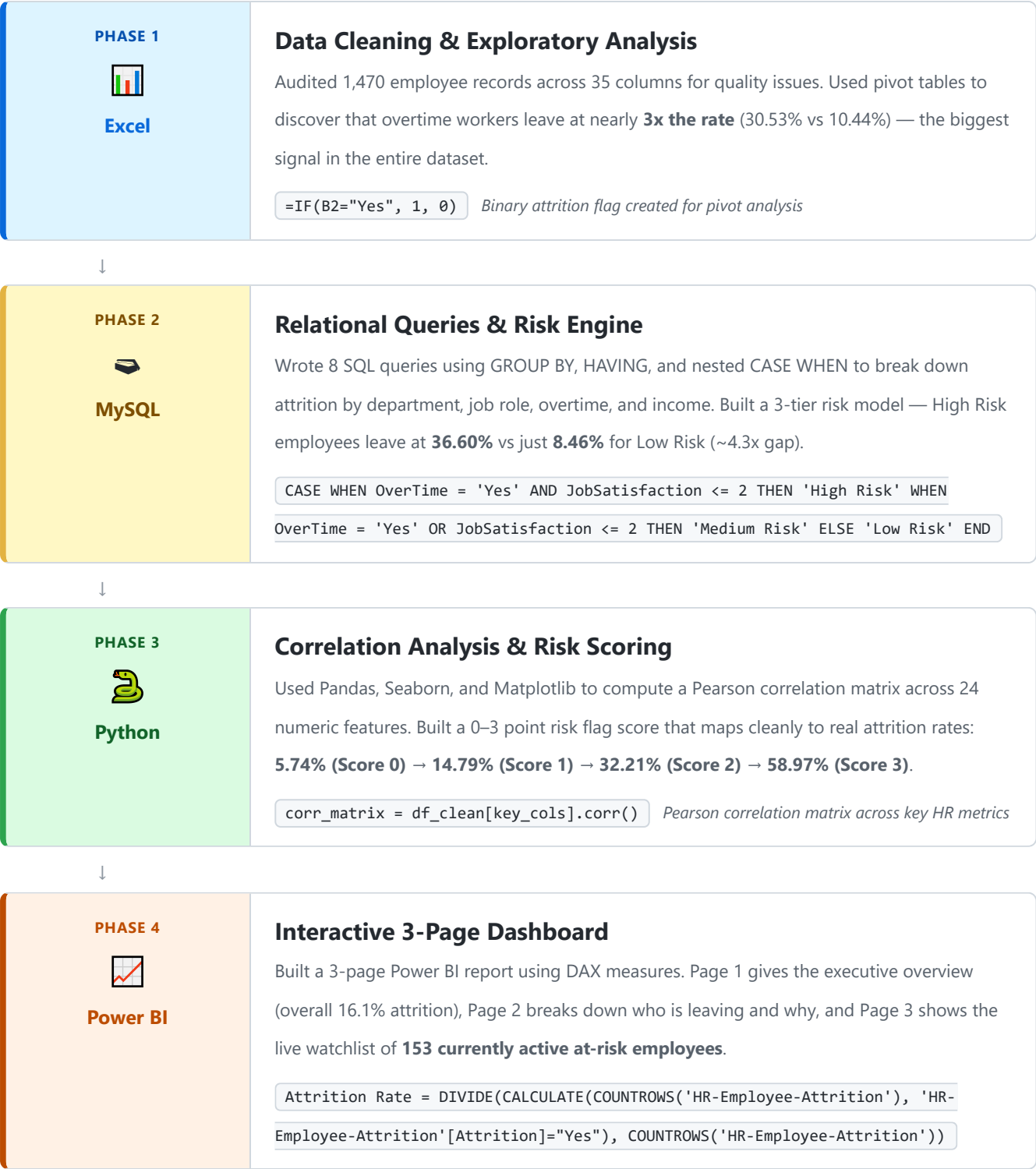
Page 2: Deep Dive



Page 3: Risk Watchlist



Project Architecture & Analytics Lifecycle





Strategic Recommendations for HR Leadership

Strategic Pillar	Identified Root Cause	Proposed HR Intervention	Expected Business ROI
1. Stop Too Much	People working overtime quit at a	Put a strict limit on how much overtime people can	Fewer people will quit from burnout, which

Strategic Pillar	Identified Root Cause	Proposed HR Intervention	Expected Business ROI
Overtime	30.5% rate (nearly 3 times more than normal).	work in a row, and make sure they get enough rest.	saves us money on hiring replacements.
2. Help New Hires Better	About 30% of all people who quit leave within their first 2 years.	Managers should talk to new hires at 30, 60, and 90 days to see how they are feeling and catch problems early.	New people will be happier and stick around longer instead of leaving right away.
3. Fix Unfair Pay	People who quit were making \$2,046 less per month than those who stayed.	Check the salaries of our lowest-paid workers (especially Sales Reps and Lab Techs) and pay them a fair wage.	Stops our best people from leaving just to get a better paycheck somewhere else.
4. Talk to At-Risk Staff Now	We found 153 current employees who are at very high risk of quitting right now.	HR needs to sit down and talk to these 153 people immediately to find out what's bothering them and try to fix it.	We keep our experienced staff and don't have to scramble when they suddenly quit.

Repository Directory Structure

```

HR-Attrition-Workforce-Intelligence/
├── README.md                <-- Executive case study & business r
├── LICENSE                  <-- MIT License
├── .gitignore               <-- Ignores temp Excel (~$*), Python
├──
├── assets/                  <-- Visual assets rendered in README
│   ├── dashboards/
│   │   ├── page1_executive_overview.png
│   │   ├── page2_deep_dive.png
│   │   └── page3_risk_watchlist.png
│   └── python_visuals/
│       ├── 01_attrition_overview.png
│       ├── 02_jobrole_satisfaction.png
│       └── 03_correlation_heatmap.png

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├── 04_risk_score_validation.png
├── data/
│   ├── raw/
│   │   └── HR-Employee-Attrition.csv    <-- Original raw dataset (1,470 recor
│   └── processed/
│       ├── dept_attrition.csv           <-- Departmental summary
│       ├── dept_attrition_python.csv    <-- Departmental summary (Python)
│       ├── dept_gender_attrition.csv    <-- Cross-demographic metrics
│       ├── hr_data_with_risk_score.csv  <-- Full dataset with Python risk scores
│       ├── income_comparison.csv        <-- Salary comparison metrics
│       ├── jobrole_attrition.csv        <-- Role-level matrix
│       ├── overtime_attrition.csv       <-- Overtime breakdown
│       ├── risk_score_validation.csv    <-- Python score validation
│       ├── risk_segments.csv           <-- SQL risk segmentation results
│       ├── watchlist_employees.csv      <-- 23 high-priority active intervent
│       └── watchlist_from_python.csv    <-- High-priority active interventions (Python)
├── excel/
│   └── HR_Attrition_Analysis.xlsx       <-- Excel workbook with formulas & pi
├── power_bi/
│   └── HR_Attrition_Visualization.pbix  <-- Interactive 3-page Power BI dashb
├── python/
│   ├── HR_Attrition_Employee_Data_Analyst_Project.ipynb <-- Jupyter notebook for EDA & Risk Scoring
│   └── requirements.txt                     <-- Python environment dependencies
└── sql/
    ├── 01_schema_setup.sql               <-- Database creation & table schema
    ├── 02_analytical_queries.sql         <-- Core analytical queries (GROUP BY
    └── 03_risk_segmentation_watchlist.sql <-- Nested CASE WHEN & active wat

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Author & Contact

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If your team is hiring, or if you'd just like to chat about this project or data analytics in general, I would love to connect!



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★ If you found this project interesting, a star on the repository is greatly appreciated!